

IMS AXES

A PROJECT REPORT

Submitted by

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220170107055

In partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER ENGINEERING

VISHWAKARMA GOVERNMENT ENGINEERING

COLLEGE CHANDKHEDA, AHMEDABAD



Gujarat Technological University, Ahmedabad

April, 2026



Vishwakarma Government Engineering College
Chandkheda, Ahmedabad

CERTIFICATE

This is to certify that the project report submitted along with the project entitled **IMS AXES** has been carried out by **Khunt Arpilkumar Sanjaybhai (220170107055)** under my guidance in partial fulfilment for the degree of Bachelor of Engineering in Computer Engineering, 8th Semester of Gujarat Technological University, Ahmadabad during the academic year 2025-26.

Prof. Jaimin M. Shroff
Internal Guide

Prof. Kajal S. Patel
Head of the Department



GUJARAT TECHNOLOGICAL UNIVERSITY

CERTIFICATE FOR COMPLETION OF ALL ACTIVITIES AT ONLINE PROJECT PORTAL

B.E. SEMESTER VIII, ACADEMIC YEAR 2024-2025

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This is to certify that, *Aghera Krutik Dipakbhai* (Enrolment Number - 210170107017) working on project entitled with *AXES AUTOMATION* from *Computer Engineering* department of *VISHWAKARMA GOVERNMENT ENGINEERING COLLEGE, CHANDKHEDA* had submitted following details at online project portal.

Internship Project Report	Completed
---------------------------	-----------

Name of Student : A g h e r a K r u t i k
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Name of Guide : Mr.Dave Nakul Rajnikant

Signature of Student : _____

*Signature of Guide : _____

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To Whomsoever It May Concern

This is to certify that Krutik Aghera is an intern in our organization from January 21, 2025, until present. During his internship, he has been actively involved in the IMS Automation.

Throughout this period, he has demonstrated punctuality, dedication, and a strong eagerness to learn. We appreciate his contributions and wish him all the very best in his future endeavors.

For Meditab Software (India) Pvt. Ltd.

Chaitra Shetty

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DECLARATION

I hereby declare that the Internship/ Project Report submitted along with the Internship entitled **IMS AXES** submitted in partial fulfillment for the degree of Bachelor of Engineering in Computer Engineering to Gujarat Technological University, Ahmadabad, is a Bonafide record of original project work carried out by me at **Meditab** under the supervision of **Nevil Patel** and that no part of this report has been directly copied from any student's reports or taken from any other source, without providing due reference.

Name of Student

ARPILKUMAR SANJAYBHAI KHUNT

Sign Student

ACKNOWLEDGEMENT

I would like to take this opportunity to extend my heartfelt appreciation to everyone who played a role in the successful completion of my internship project.

First and foremost, I am deeply grateful to my internal mentor, **Prof. Jaimin M. Shroff**, for her constant encouragement, thoughtful feedback, and valuable insights throughout this journey. Her unwavering support, guidance, and deep knowledge have been instrumental in shaping the outcome of this project.

I would also like to thank my external guide, **Nevil Patel** at **Meditab**, for his professional guidance and ongoing support during my time at the organisation. His practical suggestions and mentorship greatly enriched my learning experience.

My sincere thanks go to **Prof. Kajal S. Patel**, Head of the Department, for her encouraging words, constructive feedback, and continued motivation throughout the internship.

Lastly, I would like to acknowledge the efforts of all my faculty members and express my gratitude to my friends and peers for their cooperation and assistance, which contributed significantly to the successful completion of this project.

Thank you,

ARPILKUMAR SANJAYBHAI KHUNT

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ABSTRACT

Purpose of Internship:

The internship provided a valuable opportunity to develop real-world technical and soft skills through hands-on project work. It helped in strengthening problem-solving capabilities, communication, and collaborative skills while gaining deep insights into test automation frameworks and practices.

Scope of Internship:

*The scope included developing robust and scalable automation frameworks using tools such as **Selenium**, **TestNG**, **Maven**, **Extent Reports**, and **UFT**. The focus was on creating maintainable and reusable scripts capable of handling a wide range of testing scenarios, including both sanity and regression testing.*

Work Done:

The internship involved two primary projects:

1. IMS AXES:

Worked on the development of automation scripts for a company-internal platform named "AXES" (Automation Execution Environment for Selenium and IMS). The system automated sanity and IMS testing of internal web applications using test suites provided by the QA team. The scripts were modular and designed to ensure minimal human intervention during execution.

2. Smart Support System (Quickcap Support Bot):

Smart Support System (Quickcap Support Bot) is an AI-powered application built using a Retrieval-Augmented Generation (RAG) approach to deliver accurate and context-aware responses. The system consists of two main modules: an ingestion pipeline that processes PDFs and other resources to build a dynamic knowledge base, and a conversational interface for real-time user interaction. It leverages streaming capabilities via Streamlit to provide responses word-by-word, enhancing user experience with minimal latency.

Together, both projects contributed to a significant reduction in manual testing efforts and knowledge retrieval, increased test coverage, and enhanced the maintainability of test suites - thereby improving overall product quality and testing efficiency.

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Abbreviations

Abbreviation	Full Form
API	Application Programming Interface
SQL	Structured Query Language
IMS	Intelligent Medical Software
AXES	Automation Export Environmental Software
EHR	Electronic Health Record
IDE	Intelligent Development Environment
JDK	Java Development Kit
QA	Quality Assurance

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Chapter 1: Overview of the Company

This chapter will introduce the company where I completed my internship. Following the company overview, I will describe the training and learning experiences during the internship period. Each chapter in this report covers a dedicated topic that I explored during my training, supported by real-world examples and practical implementations. At the end of each chapter, I have highlighted the key outcomes for quick recall. This chapter marks the beginning of the journey by providing an introduction to Meditab Software Pvt. Ltd.



Figure 1.1 Company Logo

About the Company

Meditab Software Pvt. Ltd. is a private software solutions company specialising in healthcare technology. Founded in **1998**, Meditab's mission is to provide advanced, intuitive solutions that help healthcare providers improve patient care and streamline their practice operations. The company is well known for its flagship product, **Intelligent Medical Software (IMS)**, a comprehensive multispecialty EHR (Electronic Health Record) and practice management system.

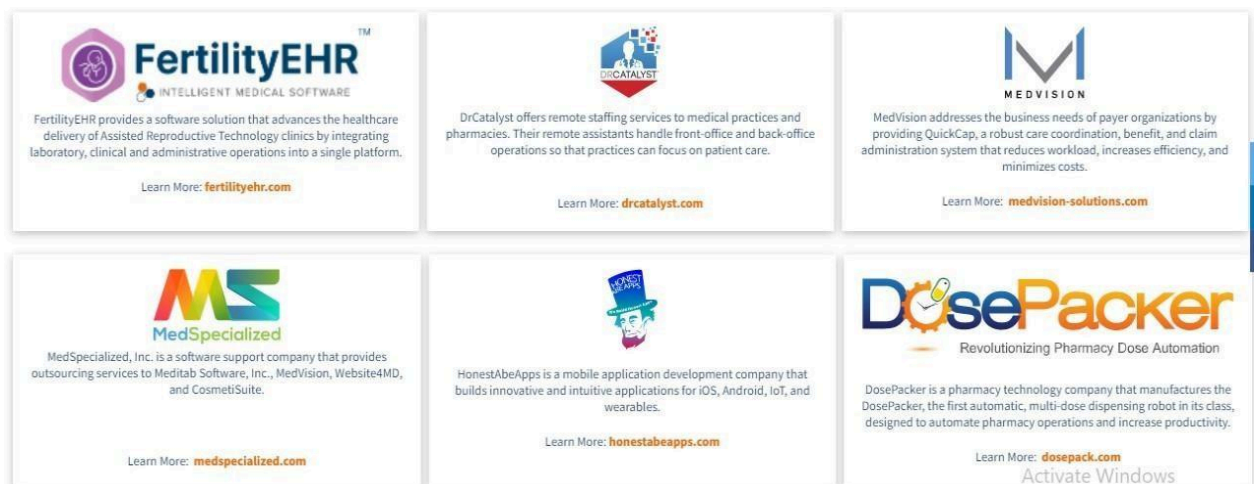


Figure 1.2 Company Product

Key Offerings

- **IMS Platform**

Includes practice management, billing, patient engagement, and telemedicine features.

- **Specialty EHRs**

Customised for over 40 medical specialities, including:

- Allergy & Immunology
- Fertility
- Ophthalmology
- ENT, Dermatology, and more

- **AXES Automation Platform**

A Selenium n-based internal tool used for automated testing, script execution, and UI tracking.

- **Customisation & Cloud Access**

Highly adaptable software with mobile and cloud capabilities.

- **Patient Education Tools**

Provides patients with condition-specific educational content to support their care journey.

Company Growth and Recognition

- **Global Offices in:**

- United States (Headquartered in Sacramento, CA)
- India
- Philippines
- Argentina

- **Recognitions**

- ISO 9001:2015 certified
- Listed in Black Book Rankings
- Winner of the 2021 Best of Sacramento Award

Scope of Work

Meditab delivers a wide range of healthcare software solutions, including:

- Multispecialty EHR systems
- Billing and practice management tools
- Telemedicine and patient engagement apps
- Automation testing frameworks (e.g., AXES)

Organization Chart

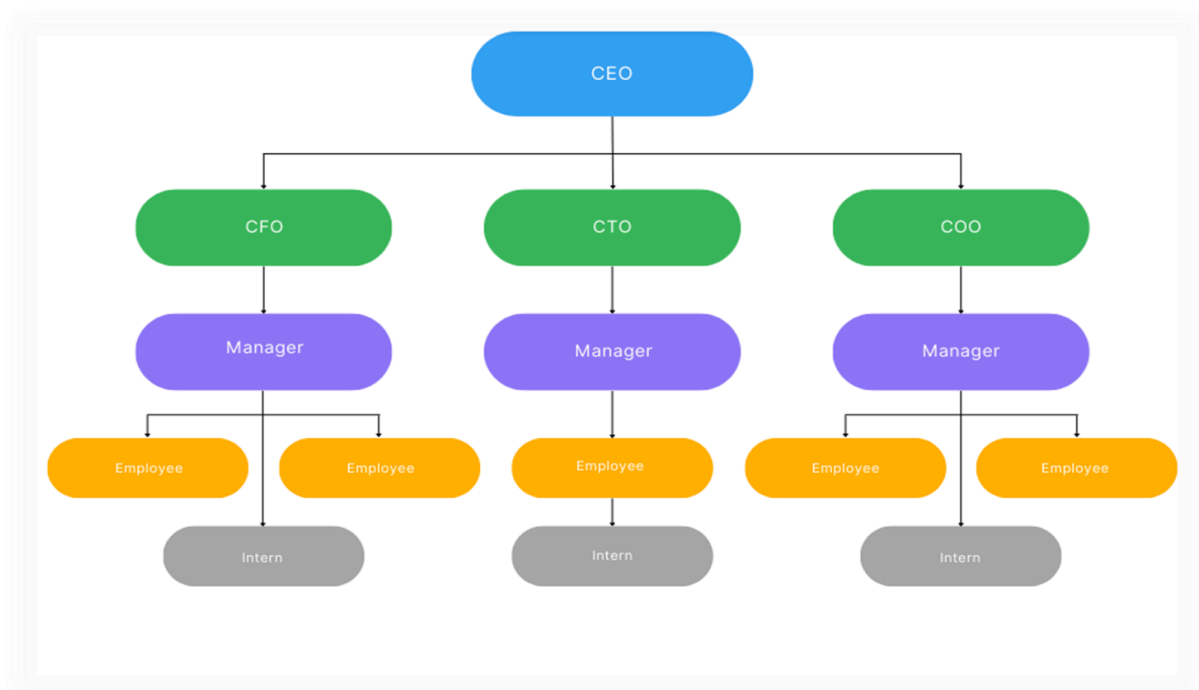


Figure 1.3 Organization Chart

Capacity Of Plant

Currently our company holds over 2600 employees. But as the company is growing rapidly its capacity is going higher and higher.

Chapter 2: Overview of Internship

2.1. Internship Summary and Project Detail

During my internship at Meditab Software Pvt. Ltd., I was provided with the opportunity to work on real-time software automation under the guidance of experienced professionals. The primary focus of the internship was the development of a dynamic automation testing framework named **IMS AXES**. This framework was aimed at streamlining and enhancing the internal QA processes by automating major workflows of Meditab's web applications.

The internship was thoughtfully structured, offering me well-rounded technical exposure across different domains:

- **SQL Training:** I learned how to work with structured databases, write complex queries, perform joins, filtering, and aggregate operations using SQL. This helped in understanding backend data validation during test automation.
- **Java Training:** I was introduced to object-oriented programming concepts in Java, learning class structures, exception handling, file handling, and API integration. Java formed the backbone of our test scripts in the AXES framework.
- **Selenium & Automation Training:** In-depth training in Selenium WebDriver enabled me to write scripts for browser-based automation. I also explored concepts like dynamic element handling, waits, cross-browser compatibility, and framework integration with TestNG and Maven.
- **UFT Framework Training :** The UFT (Unified Functional Testing) Framework Training provided hands-on experience with this enterprise-level automation tool, focusing on leveraging its object repository and checkpoint capabilities. This training was essential for understanding how to integrate diverse testing tools to create a comprehensive and robust hybrid automation framework.

- **DevOps Training** : DevOps Training provided foundational knowledge in continuous integration and continuous deployment (CI/CD) practices. This included hands-on experience setting up a complete CI/CD pipeline using **GitLab Runner**. A key focus was on containerization, utilizing **Docker** to package applications, and orchestration using **Kubernetes** to manage and deploy these containers at scale in an automated manner.
- **Langchain & LangGraph Training** : This training focused on leveraging large language models (LLMs) to build sophisticated, data-aware applications. We learned how to use **Langchain** for chaining multiple components (like prompts, LLMs, and external data sources) and **LangGraph** for defining complex, stateful LLM workflows and agentic systems, enabling more reliable and controlled interactions.

This combined training enabled me to understand the full software testing lifecycle and contribute actively to the AXES Automation project.

2.2 Project Overview

The IMS AXES project focuses on the development of a dynamic and reusable test automation framework for internal web applications and window applications used at Meditab. The goal is to reduce repetitive manual testing tasks by automating key application workflows such as login, task management, and verification processes. This system ensures that each deployment or update of the application can be validated quickly with minimal effort using automated scripts. As a result, it enhances the overall product quality and helps provide a seamless user experience to both internal stakeholders and clients.

2.3 Purpose of Project

The primary purpose of this project is to improve testing efficiency and depth through reusable, robust automation scripts. It aims to reduce time spent on repetitive testing cycles and minimise human error. By implementing a reliable automation suite, the project ensures consistency across test runs and builds a professional image in front of the client by delivering a smooth and error-free experience.

2.4 Scope of Project

This automation framework can be applied to test any web-based or desktop-based system using a dynamic, centrally-managed scripting approach. IMS AXES is scalable, allowing multiple test cases and modules to be handled by a single script logic that adapts to different UI flows.

The framework supports:

- **IMS Testing** - Basic validation to ensure stable builds
- **Regression Testing** - Comprehensive testing after updates
- **Cross-module Reusability** - One script across similar workflows
- **Batch Execution** - Automating multiple test runs together

AXES is adaptable and client-agnostic, making it viable for internal and external systems.

2.5 Objective

Two major causes of software failures are poor user experience and inadequate testing. Applications that are not rigorously tested can suffer performance issues, bugs, or even data breaches. IMS AXES aims to address these problems by introducing a highly modular and scalable automation suite.

Key objectives include:

- Improve test coverage with minimal human intervention
- Provide quick feedback loops during CI/CD deployments
- Ensure application reliability under varied scenarios
- Reuse test scripts across different clients and modules

The project also focuses on reproducibility and flexibility. Once the framework is in place, it can be updated with minimal changes to support new test cases.

2.6 Technology And Tools

- **Programming Language:** Java, JavaScript
- **Tools & Frameworks:**
 - **Eclipse IDE:** For writing and managing the codebase
 - **UFT:** For windows applications automation
 - **Maven:** For dependency management and project structure
 - **TestNG:** For designing test suites and generating reports
 - **Extent Reports:** For detailed execution reporting with pass/fail logs
 - **Web Development:** React.js, Express.js, Node.js
 - **DataBase:** PostgreSQL

These tools are chosen for their reliability, strong community support, and scalability in enterprise-level automation environments.

IMS AXES design supports modularity, dynamic object handling (e.g., dropdowns, calendars), and easy integration into existing DevOps pipelines.

Chapter 3: Project Structure and Management

3.1 Project Planning

3.1.1 Project Development Approach

The IMS AXES project followed a structured, iterative, and incremental development approach. It was divided into the following key phases to ensure effective delivery, quality assurance, and long-term maintainability:

Phase 1 – Initiation and Strategy Setup

- Define testing strategy and scope
- Gather automation feasibility and system requirements
- Identify manual components to convert to automation
- Select automation tools: **Selenium, UFT, Jenkins**, etc.

Phase 2 – Requirements Analysis and Test Planning

- Define testing requirements:
 - Unit Testing
 - Functional Testing
 - Interface Testing
 - Security and Compliance
 - Platform Compatibility
 - Performance Benchmarks
- Understand globalization and accessibility needs
- Estimate development efforts and pricing
- Obtain planning approvals and scheduling sign-offs
- Establish communication between development and QA teams

Phase 3 – Knowledge Transfer and Scenario Design

- Application demonstration and feature walkthrough
- Domain-specific knowledge transfer
- Study user/operations manuals
- Draft test plans and define test cases for key scenarios

Phase 4 – Environment and Platform Setup

- Provisioning of required hardware/software resources
- Setup and configuration of test environments
- Define execution platforms for global testing
- Design and build test suites

Phase 5 – Script Design and Automation Workflow

- Develop test architecture and script framework
- Identify reusable components (object repository, test data)
- Build test cases aligned with the designed workflows
- Execute performance scripts for worst-case scenarios
- Perform repeated test runs with varying data

Phase 6 – Execution and Final Delivery

- Execute full test suites and validate results
- Log all test outcomes with pass/fail criteria
- Track issues and update automation scripts

Phase 7 – Maintenance and Continuous Improvement

- Improve test coverage and reduce script execution time
- Add new test cases based on feature updates
- Maintain modularity and reusability for future automation efforts

3.1.2 Project Efforts and Cost Estimation

To estimate the cost and effort required for this project, we applied the **COCOMO (Constructive Cost Estimation Model)**. This model helps calculate project effort based on the complexity and size of the software in terms of **Function Points (FP)** and **Lines of Code (LOC)**.

COCOMO Classification:

The AXES Automation framework falls under the “**Application**” project category in the COCOMO model due to its purpose of enhancing QA through automation tools.

Function Point Analysis:

Parameter	Count	Simple	Average	Complex	FP Value
No. of User Inputs	22	3	4	6	88.5
No. of User Outputs	5	4	5	7	25.0
No. of Inquiries	3	3	4	6	12.0
No. of Files	8	7	10	15	80.0
External Interfaces	2	5	7	10	14.0

Table 3.1 – Function Point Calculation for COCOMO Model

Question	0	1	2	3	4	5
1. Does the system require reliable backup and recovery?	☐	☐	☐	☐	☐	☒
2. Are data communications required?	☐	☐	☐	☐	☒	☐
3. Are there distributed processing functions?	☐	☐	☒	☐	☐	☐
4. Is performance critical?	☐	☐	☐	☐	☐	☒
5. Will the system run in an existing, heavily utilized operational environment?	☐	☐	☐	☐	☐	☒
6. Does the system require on-line data entry?	☐	☐	☐	☐	☒	☐
7. Does the on-line data entry require the input transaction to be built over multiple screens or operations?	☐	☐	☐	☒	☐	☐
8. Are the master file updated on-line?	☐	☐	☐	☐	☒	☐
9. Are the inputs, outputs, files, or inquiries complex?	☐	☐	☐	☐	☒	☐
10. Is the internal processing complex?	☐	☐	☐	☒	☐	☐
11. In the code designed to be reusable?	☐	☐	☐	☐	☐	☒
12. Are conversion and installation included in the design?	☐	☐	☐	☐	☒	☐
13. Is the system designed for multiple installations in different organizations?	☐	☐	☐	☐	☐	☒
14. Is the application designed to facilitate change and ease of use by the user?	☐	☐	☐	☐	☐	☒
Total						
58.00						

Fig 3.1 – Questions Dependent on the Project

The Function Points is:

Fig 3.2 – Function Points Calculation

Programming Language	LOC/FP (average)	Select
Assembly Language	320	☐
C	128	☐
COBOL	105	☐
Fortran	105	☐
Pascal	90	☐
Ada	70	☐
Object-Oriented Languages	30	☒
Fourth Generation Languages (4GLs)	20	☐
Code Generators	15	☐
Spreadsheets	6	☐
Graphical Languages (icons)	4	☐

LOC/FP:

Software Project	a _b	b _b	c _b	d _b	Select
Organic	2.4	1.05	2.5	0.38	☐
Semi-detached	3.0	1.12	2.5	0.35	☐
Embedded	3.6	1.20	2.5	0.32	☒

$$\text{Effort (E)} = a_b(KLOC)^{b_b} = 44.18 \quad \text{Duration (D)} = c_b(E)^{d_b} = 8.40$$

Fig 3.3 – LOC/FP Calculation Diagram

3.1.3 Roles and Responsibilities

Name

Arpil Sanjaybhai Khunt

Role

Requirement Gathering,
Script Design, Test
Execution, Reporting,
Designing, Documentation

3.2 Project Scheduling and Representation

Project scheduling was executed using a **Gantt Chart**, a type of bar chart used to represent the timeline of tasks in a project. Each horizontal bar in the chart reflects the start and end date of a specific activity, allowing stakeholders to track project progress and dependencies visually.

The Gantt chart covered:

- Training and Requirement Phase
- Test Plan Design and Tool Setup
- Script Development
- Testing and Debugging
- Reporting and Documentation

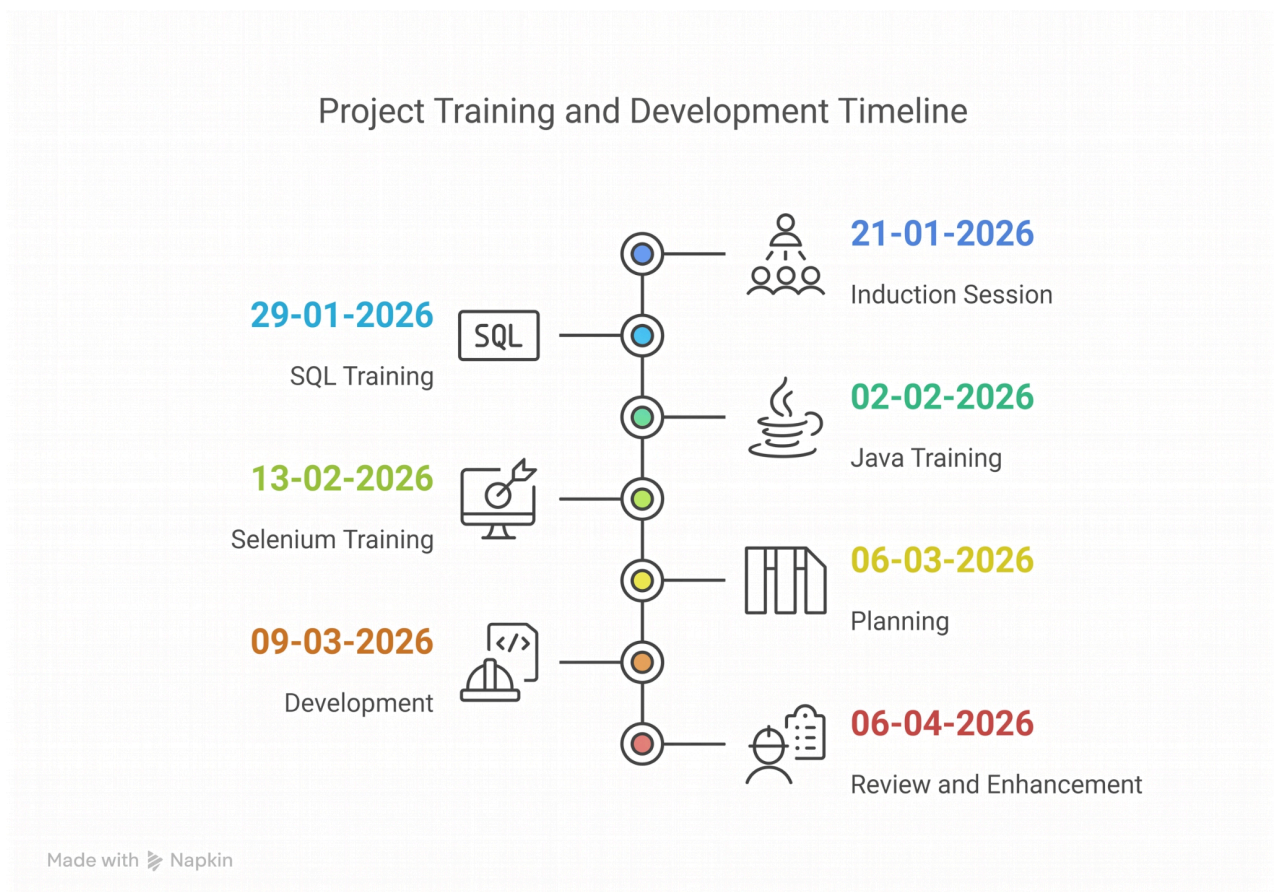


Fig 3.4 – Gantt Chart Representation of IMS AXES Project

Chapter 4: Project And System Requirement Study

4.1 User Characteristics

The IMS AXES project is developed for a range of user types involved in the software lifecycle. Each type of user interacts with the system from a different perspective and for different goals:

- **Hardware Developers:**
These users are responsible for validating the functional requirements of the hardware integrations (if any) and ensuring that system compatibility is maintained.
 - **Production Team:**
This team uses the automation framework for end-to-end product testing before deployment. Their main objective is to ensure that the product meets all defined quality standards.
 - **Sales and Marketing Team:**
They use the tested and stable application to conduct demonstrations and deliver reliable showcases to prospective clients. A well-tested app adds confidence in client presentations.
 - **Clients and End Users:**
Clients benefit from a fully customized and tested version of the product. They rely on a bug-free and efficient application as part of their business operations. Their feedback helps in iterative testing and improvements.
-

4.2 Hardware And Software Requirements

To run the AXES Automation Framework smoothly, a predefined set of hardware and software resources are required. These requirements ensure optimal performance and support across environments.

Software Requirements:

- **Operating System:** Windows 10/11 or compatible Linux distribution
- **Development IDE:** Eclipse IDE (or any Java-compatible IDE)
- **Java (JDK):** Version 17 or higher
- **UFT:** Window application automation tool
- **TestNG:** For structuring test suites and reporting test execution
- **Maven:** For dependency management and build automation
- **Extent Reports:** For enhanced HTML-based test reporting
- **Browser Drivers:** ChromeDriver, GeckoDriver, etc.

Hardware Requirements:

- **Processor:** Minimum 3.5 GHz (Quad Core or higher preferred)
 - **RAM:** Minimum 8 GB (16 GB recommended for smoother execution of parallel test suites)
 - **Storage:** At least 512 GB SSD for efficient read/write performance
 - **Display:** Full HD (1920x1080) recommended for UI validation tasks
 - **Network:** Reliable internet for downloading dependencies and executing cloud-based tests (if applicable)
-

4.3 Assumptions and Dependencies for Projects

Assumptions:

- No other heavy applications are running on the system during test execution.
- The system running tests has administrative privileges.
- Auto-lock, screen saver, or sleep mode is disabled during script execution.
- Browser drivers are pre-installed and properly linked in the system PATH.
- The system has uninterrupted internet access for Maven and WebDriver updates.

Dependencies:

- **Reusable Function Libraries:**
 - Core logic components used throughout multiple test modules
 - Handlers for dynamic dropdowns, calendars, modal dialogues, etc.
- **Main Test Scripts:**
 - Calls to reusable functions and modules for test scenarios
 - Structured around the Page Object Model (POM)
- **Test Data Files:**
 - Excel/CSV-based parameter files used for data-driven testing
 - Environment-specific data (staging, dev, production)
- **Configuration Files:**
 - config.properties for dynamic environment settings
 - Logging and reporting setup
- **Report Viewers and Log Analysers:**
 - Tools like Extent Reports or custom HTML reports to analyse results
- **Google Looker Studio:**
 - It's a free tool that turns your data into informative, easy-to-read, and shareable dashboards and reports. It allows users to connect to hundreds of data sources, visualize data insights, and make data-driven decisions.

This combination of hardware/software setup and well-defined user expectations ensures that the IMS AXES framework can be executed with high reliability and scalability across environments.